SL&I Semester 3 Final Exam Study Guide

The final exam on Nov 4, 2026 (1:00–2:30 PM) has approximately 60 single-best-answer questions with approximately 20 each on behavioral economics, survival analysis, and biostatistics. This guide is a companion to the slides, not a replacement.

Part I: Behavioral economics (Sessions 1–2)

The exam tests whether you can name the economic concept at work in a patient or policy scenario and predict its effect.

Why health insurance exists

- **Spending is uncertain, variable, and right-skewed.** Most people spend little in a year; a few spend enormous amounts. That is why saving or borrowing cannot replace insurance.
- **Insurance is financial.** It reduces the variance of income by pooling risk (consumption smoothing). It does not restore health.
- **Risk premium.** A risk-averse person's willingness to pay (WTP) = expected cost + risk premium. They buy only if the premium is at or below their WTP.
- **Premium = expected cost + loading.** The actuarially fair premium equals expected cost alone.
- **Primary vs. secondary risk.** Primary risk is the individual's chance of a costly event. Secondary risk is the insurer's risk of misjudging average costs, which shrinks as the pool grows.

Moral hazard and cost sharing

Concept	What it means	Clinical cue
Ex ante moral hazard	Coverage changes behavior that makes the event more likely	Risky behavior "because insurance will pay"
Ex post moral hazard	Coverage increases use of care after the event	Tests or visits the patient would not buy at full price
Price elasticity	Moral hazard is largest for discretionary, price-elastic care	Elective, cosmetic, or comfort services vs. insulin or dialysis
Deductible	Patient pays all costs up to a set amount	"First $1,500"
Coinsurance	Patient pays a percentage after the deductible	"20% of the remainder"
Copayment	Fixed dollar amount per service	"$25 per visit"
Stop-loss (out-of-pocket max)	Caps total patient spending	Large bills that hit the cap
Own-price effect	Higher price for a service lowers its use	Higher drug copay, fewer fills of that drug
Cross-price effect	Price of one service changes use of a related service	Higher drug copay, fewer office visits
Offset effect	Cutting needed care causes harm and higher downstream costs	Stops diuretic, then admitted for heart failure

Be able to calculate a patient's out-of-pocket cost from a deductible, coinsurance, and out-of-pocket maximum.

Supply-side controls

- **Managed care:** the insurer shapes which services are provided.
- **Gatekeeping:** a primary care referral is required before specialty care.
- **Capacity control:** the system limits supply (beds, scanners), producing waits rather than prices, as in Canada.

Risk selection

- **Adverse selection** happens when people choose coverage, know more about their own risk than the insurer, and premiums do not match individual risk. High-risk people buy; low-risk people leave.
- **The death spiral.** Premium set at the pool average, low-risk people exit, the average rises, more people exit. Be able to work a WTP table round by round.
- **Favorable selection (cream-skimming)** is the insurer's side: designing plans that attract healthy people, such as gym benefits or adverse tiering of costly drugs.
- **Remedies:** group or universal coverage, preexisting-condition rules paired with an individual mandate, risk adjustment, reinsurance, and risk corridors. Know what each does and its limits.
- **Group insurance** reduces adverse selection, spreads fixed (loading) costs across many enrollees, and adds bargaining power. Know the downside of employer-based coverage, such as job lock.

The two fundamental tradeoffs

1. **Financial protection vs. moral hazard:** more coverage protects people but raises use and cost.
2. **Competition vs. adverse selection:** rules that limit selection (standard benefits) also limit competition and choice.

Part II: Survival analysis (Sessions 3–4)

The exam tests whether you can code a patient's data, read a Kaplan-Meier curve, and interpret a hazard ratio from a trial report.

Core vocabulary

- **Survival analysis** studies time until an event ("failure"), such as death, recurrence, or relapse. Use it when patients are followed for different lengths of time.
- **Survival time** is measured from each patient's own entry (or randomization), not from the calendar start of the study.
- **Status** is coded 1 = failed (event) and 0 = censored. Be able to build and check a data table from patient stories.
- **Three goals:** estimate survival, compare survival between groups, and assess how explanatory variables relate to survival (Cox regression).

Censoring

Censoring means we have incomplete information about a patient's survival time.

Type	What happened	Example
Right	Event not yet seen when follow-up stops	Study ends, loss to follow-up, withdrawal
Administrative (right)	Study closed before the event	Late enrollee still event-free at data lock
Left	Event occurred before observation began	Already infected at first visit

Type	What happened	Example
Interval	Event occurred between two observations	Absent at Year 2 exam, present at Year 3

- **Non-informative censoring** is the key assumption: patients who leave have the same future risk as those who stay. A job relocation is non-informative.
- **Informative censoring** occurs when leaving is linked to prognosis, such as the sickest patients dropping out. It biases results; if high-risk patients leave, survival is overestimated.

Survival function and Kaplan-Meier curves

- **S(t) = P(T > t):** the probability of remaining event-free beyond time t. It starts at 1 and never rises.
- The Kaplan-Meier curve is a **step function** that drops only when events occur.
- **Tick marks** show censored patients. They leave the risk set without causing a drop.
- The curve reaches 0 only if **every** patient has the event.
- The **number-at-risk table** under the curve shows who is still being followed. A steep fall in numbers at risk with a flat curve means censoring (often administrative), not deaths.
- Estimates at the tail of the curve are **less precise** because few patients remain at risk.

Hazard function

- **h(t)** is the instantaneous rate of the event at time t among those who have survived to t. Think of it as a velocity.
- It is a **rate, not a probability**: nonnegative, with no upper bound.
- Know the five shapes and a clinical example of each:

Shape	Pattern	Example
Constant	Same risk at every time	Risk does not depend on time already survived
Increasing	Rises over time	Aging, progressive disease
Decreasing	Highest early, then falls	Complications after surgery
Bathtub	High, low, then high again	Transplant: early complications, later aging and graft failure
Unimodal	Rises to a peak, then falls	Stroke risk after a TIA

Endpoints and hazard ratios

- **Composite endpoints** such as event-free survival (EFS) count the first of several events. Read the definition: "death from any cause" includes an accidental death.
- **Overall survival (OS)** counts death from any cause.
- **Hazard ratio (HR):** the ratio of hazards between arms over follow-up. HR 0.66 means a 34% lower hazard. The null value is **1**; a 95% CI that excludes 1 is significant at $\alpha = 0.05$.
- An HR is not the proportion who survived, the proportion who died, or how much longer patients lived.

Part III: Biostatistics (Sessions 5–6)

The exam is derived from the sessions emphasizing: Is it real? Does it matter? It also asks what the data look like and how two things are related.

Is it real? Hypothesis testing, errors, and power

- $\mathbf{H_0}$ states no effect; $\mathbf{H_a}$ states an effect. Be able to write both in words for a trial.

- **p-value:** the probability of results at least this extreme **if H_0 were true**. It is not the probability H_0 is true or that the drug works.
- **Type I error (α):** rejecting a true H_0 (false positive). Patients get an ineffective drug.
- **Type II error (β):** failing to reject a false H_0 (false negative). An effective drug is abandoned.
- **Power = 1 − β.** It rises with larger n, larger true effect, less variability (paired or crossover designs), and higher α. Targets: α = 0.05, power ≥ 80%.
- **Fail to reject is not accept.** A non-significant result may mean the study was too small. Read the CI.
- **Multiple comparisons:** k tests at α = 0.05 give a $1 - 0.95^k$ chance of at least one false positive (3 tests 14%, 6 tests 26%).

Choosing the test

Outcome	Comparison	Test
Continuous	2 independent groups	Two-sample t-test
Continuous	Same patients measured twice	Paired t-test
Continuous	3+ independent groups	One-way ANOVA, then post-hoc
Categorical (yes/no)	2 or more groups	Chi-square
Continuous	Continuous; strength of linear association	Pearson correlation (r)
Continuous	Change in outcome per unit of predictor	Linear regression (β)

Start with the outcome type. "Three groups" means ANOVA only if the outcome is continuous. A significant ANOVA shows at least one mean differs, not which. Splitting a continuous predictor into groups loses power; regression keeps it.

Confidence intervals and the null value

- A 95% CI is a range of plausible values for the true effect and a measure of precision. It is not where 95% of patients fall.
- **Null value:** 0 for differences (means, ARR, β); 1 for ratios (RR, OR, HR). A CI that excludes the null means $p < 0.05$.
- **Width** depends on n and SD: $SE = SD \div \sqrt{n}$. More patients or less noise gives a narrower CI. A 99% CI is wider than a 95% CI.

Does it matter? Clinical significance

Compare the CI to the MCID (the smallest change patients or clinicians would notice as beneficial), not just to the null.

Category	CI vs. null	CI vs. MCID	Meaning
Significant and meaningful	Excludes null	Point estimate at or beyond MCID	Real and worth acting on
Significant, not meaningful	Excludes null	Entirely short of MCID	Real but too small to matter
Inconclusive	Includes null	Includes MCID	Can't rule out a meaningful effect

Category	CI vs. null	CI vs. MCID	Meaning
Convincingly negative	Includes null	Excludes MCID	Any true effect is too small to matter

- Use **absolute measures**: ARR = control risk − treatment risk; NNT = 1 ÷ ARR, always with a time frame. A large RRR can hide a tiny ARR.
- **Within-arm change is not a treatment effect.** The effect is the difference between arms. Control groups improve too (regression to the mean, attention, Hawthorne effect, changes in usual care).

What do the data look like?

- **Shape decides the summary.** Symmetric: mean = median, report mean and SD. Positive (right) skew: mean > median, report median and IQR (length of stay, triglycerides, CRP). Negative skew: mean < median.
- **Normal curve:** 68% within ±1 SD, 95% within ±2 SD, 99.7% within ±3 SD.
- **z-score** = (value − mean) ÷ SD.
- **Reference range** = central 95% of healthy people, so 5% of healthy people fall outside it. With 20 tests, about 64% of healthy people have at least one flag.
- **SD vs. SE:** SD describes spread of patients and does not shrink with n. SE describes precision of the mean and shrinks with n. SE error bars make data look tidier than the patients are.

How are two things related?

- **r** runs from −1 to +1. Sign = direction; size = how tightly points hug a straight line. **r^2** = share of variation explained (r = 0.3 gives 9%).
- **Four ways r misleads:** correlation is not causation; r ≈ 0 can hide a curved (J-shaped) relationship; one outlier can make or break r; r is not the slope. Use Spearman ρ for skewed or ordinal data or outliers.
- **Linear regression slope (β):** the difference in mean outcome per one-unit change in the predictor. Predict only within the range of the data; beyond it is extrapolation.
- **R^2** is the share of variation the model explains. A large study can find a highly significant association with a small R^2.
- **Adjusted vs. unadjusted:** an adjusted β holds other variables constant. A large change after adjustment means confounding. Regression adjusts only for what was measured; observational data cannot exclude residual confounding.

Outcome	Model	Reports	Null
Continuous	Linear regression	β (difference in mean per unit)	0
Binary	Logistic regression	Odds ratio	1
Time-to-event	Cox proportional hazards	Hazard ratio	1

When the outcome is common, the OR overstates the relative risk. Cross-sectional data show association, not causation.

Landmark studies from the slides

Exam vignettes are built on the studies used in class. For each one, find it in the slides and be able to say what it showed and which concept it illustrates.

Study	Session	What it showed	Concept it illustrates
Tamblyn et al., JAMA 2001	1–2	After coinsurance and deductibles were added for elderly and welfare	Own-price and offset effects; cost sharing

Study	Session	What it showed	Concept it illustrates
(Quebec drug cost sharing)		recipients, use of both essential and less-essential drugs fell. Cutting essential drugs was linked to more serious adverse events and ED visits, in a dose-response pattern.	assumes capacity to pay and rational choices
Jacobs & Sommers, NEJM 2015 (HIV drug tiering)	1–2	Some ACA marketplace plans placed all HIV drugs, even generics, on the highest cost-sharing tier.	Adverse tiering as favorable risk selection through benefit design
KEYNOTE-522 (pembrolizumab in early triple-negative breast cancer)	3–4	Adding pembrolizumab to neoadjuvant chemotherapy improved event-free survival and, with longer follow-up, overall survival.	Composite endpoints (EFS), Kaplan-Meier curves, number at risk, censoring, hazard ratios
DASH-Sodium (randomized feeding trial)	6	Lowering dietary sodium lowered blood pressure under controlled conditions.	Randomized evidence supports a causal claim that a cross-sectional regression cannot make

The slides also use teaching examples you should be able to rework without notes:

- **Ms. D** (Sessions 5–6): a news story with $p < 0.001$ in 24,000 adults; one flagged lab out of 20; how much halving salt would lower her BP.
- **Veltrazide** (Session 5, fictional): writing H_0 and H_a, Type I and II errors, power and CI width, and an inconclusive result whose CI spans 0 and the MCID.
- **SALT-Check** (Session 6, fictional): reading unadjusted and adjusted regression output, R^2, extrapolation, and odds ratios.

Formula sheet and common traps

You may use a basic calculator. Practice each formula by hand until it is automatic.

Formula	Use it to
Out-of-pocket = deductible + coinsurance % × (bill − deductible), capped at the out-of-pocket max	Find what a patient pays
WTP = expected cost + risk premium	Decide whether a person buys a policy
Premium = expected cost + loading	Compare a premium to WTP
Survival time = event or censoring time − entry time	Code a patient's data
$z = (\text{value} - \text{mean}) \div \text{SD}$	Judge whether a value is unusual
$SE = SD \div \sqrt{n}$	Find the precision of a mean
95% CI ≈ mean ± 1.96 × SE	Build a CI for a mean
ARR = control risk − treatment risk	Measure absolute benefit

Formula	Use it to
RRR = ARR ÷ control risk	Measure relative benefit
NNT = 1 ÷ ARR (with a time frame)	Translate benefit for patients
P(at least one false positive) = $1 - 0.95^k$	Multiple tests or lab panels
r^2 = share of variation explained	Interpret a correlation
Predicted change = β × change in predictor	Use a regression slope

Common traps:

- Calling a CI significant because it "doesn't include 0" when the measure is a ratio. Check the null value first.
- Reading "fail to reject" as "no effect." Read the CI against the MCID.
- Reporting RRR to a patient instead of ARR and NNT.
- Picking ANOVA because there are three groups when the outcome is yes/no.
- Treating r as a slope, or a small p-value as proof of causation.
- Treating a hazard as a probability, or an HR as the share who survived.
- Counting an accidental death as censored when the endpoint includes death from any cause.
- Using a regression to predict outside the range of the data.
- Calling within-arm improvement a treatment effect.
- Confusing ex ante with ex post moral hazard, or own-price with cross-price effects.

Self-check before the exam

Tick each item only when you can do it without notes. If you can't, go back to the slides for that session.

Behavioral economics

- ☐ Explain why saving cannot replace insurance, using the shape of health spending
- ☐ Classify a scenario as ex ante or ex post moral hazard
- ☐ Calculate out-of-pocket cost with a deductible, coinsurance, and out-of-pocket max
- ☐ Tell own-price, cross-price, and offset effects apart in a vignette
- ☐ Work a death-spiral table for two rounds
- ☐ Match each remedy (mandate, risk adjustment, reinsurance, risk corridors) to the problem it solves
- ☐ State both fundamental tradeoffs with an example
- ☐ Summarize Tamblyn 2001 and Jacobs & Sommers 2015

Survival analysis

- ☐ Code survival time and status from patient stories
- ☐ Name the censoring type in a scenario and say whether it is informative
- ☐ Interpret S(t) at a given time and explain the step, tick marks, and number at risk on a Kaplan-Meier curve
- ☐ Explain why a hazard can exceed 1 and match a clinical pattern to its hazard shape
- ☐ Decide whether a given death counts as an event for EFS or OS
- ☐ Interpret an HR and its CI in one sentence

Biostatistics

- ☐ Write H_0 and H_a and describe who is harmed by a Type I vs. Type II error
- ☐ Predict how n, SD, α, and target effect change power and CI width

- ☐ Pick the right test from the outcome type and the comparison
- ☐ Interpret a p-value and say whether a CI crosses the correct null value
- ☐ Classify a result into one of the four significance categories using the MCID
- ☐ Calculate ARR, RRR, and NNT and say the result in words a patient would understand
- ☐ Choose mean/SD or median/IQR from a distribution's shape; calculate a z-score
- ☐ Tell SD from SE and calculate both ranges for a sample
- ☐ Interpret r, r^2, a regression slope, R^2, and adjusted vs. unadjusted estimates
- ☐ Match linear, logistic, and Cox regression to the outcome and name each one's null value
- ☐ Answer both of Ms. D's questions in one sentence each